

Chemical Identity is Not Causation

Molecules, Context, and the False Grammar of Bodily Explanation.

Why “my cortisol is high,” “MSG causes headaches,” and “my hormones made me do it” are structurally flawed explanations—and how to build a better biological filter.



The Substantial vs. Situated Error



Chemical Essentialism

It's a chemical, therefore dangerous.



Endogenous Innocence

It's natural, therefore safe.



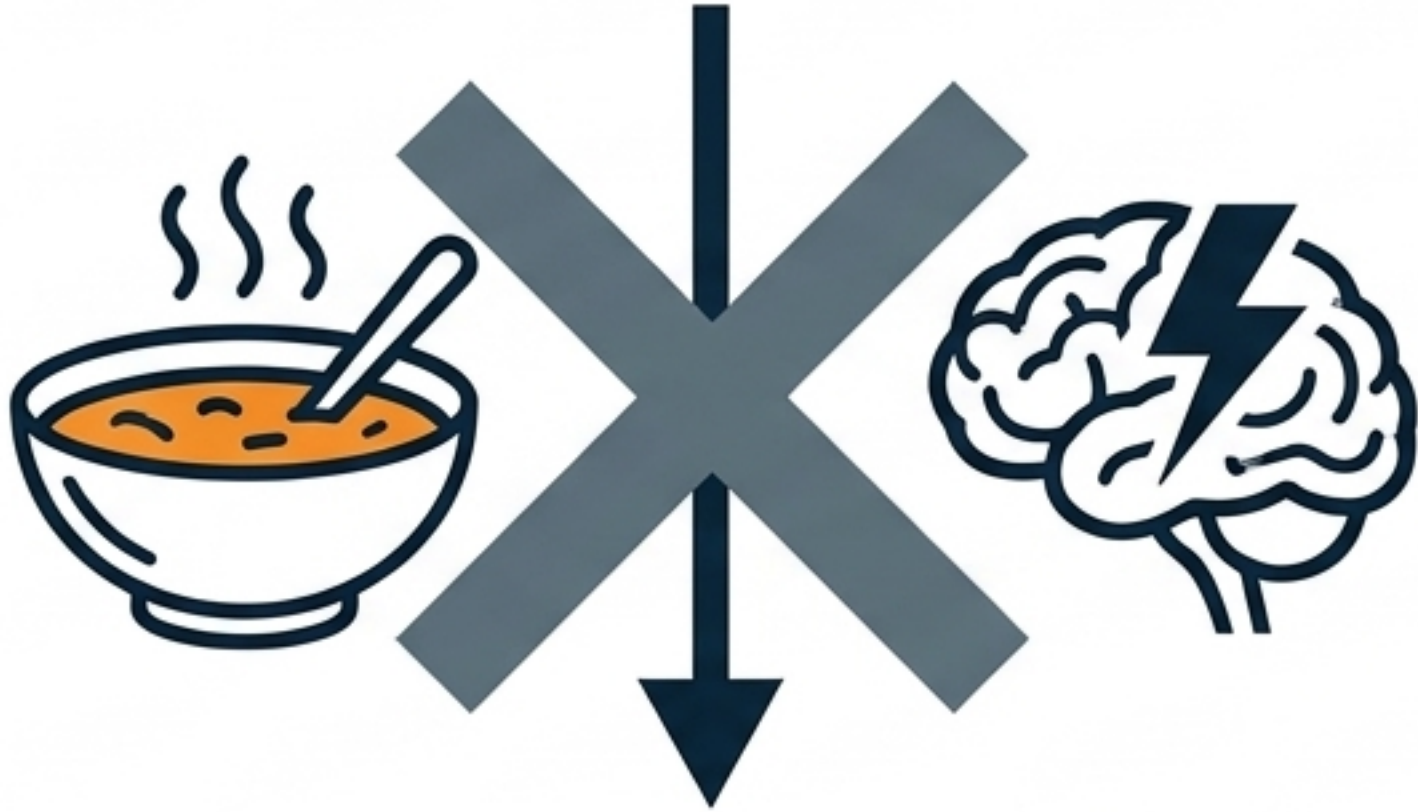
Situated Causation The Correct View

An effect is a momentary relation among material identity, exposure, pathway, organism, and time.

We routinely mistake what a substance is for what a substance does in context.

The Principle of Causal Nonidentity

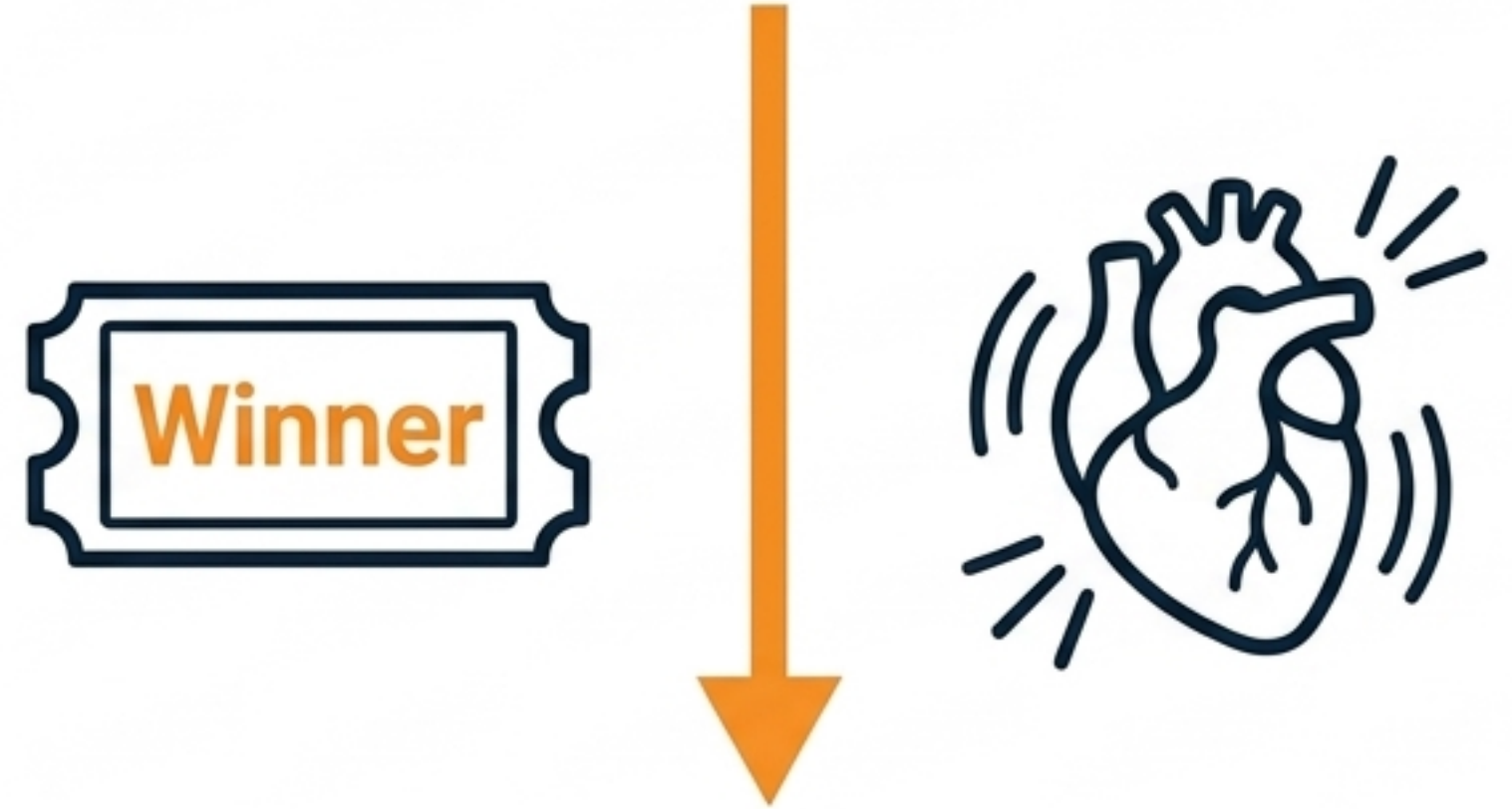
The Real Chemical



**You eat MSG
You get a headache.**

(A real chemical exposure, but false attribution to the chemical itself).

The Fake Event

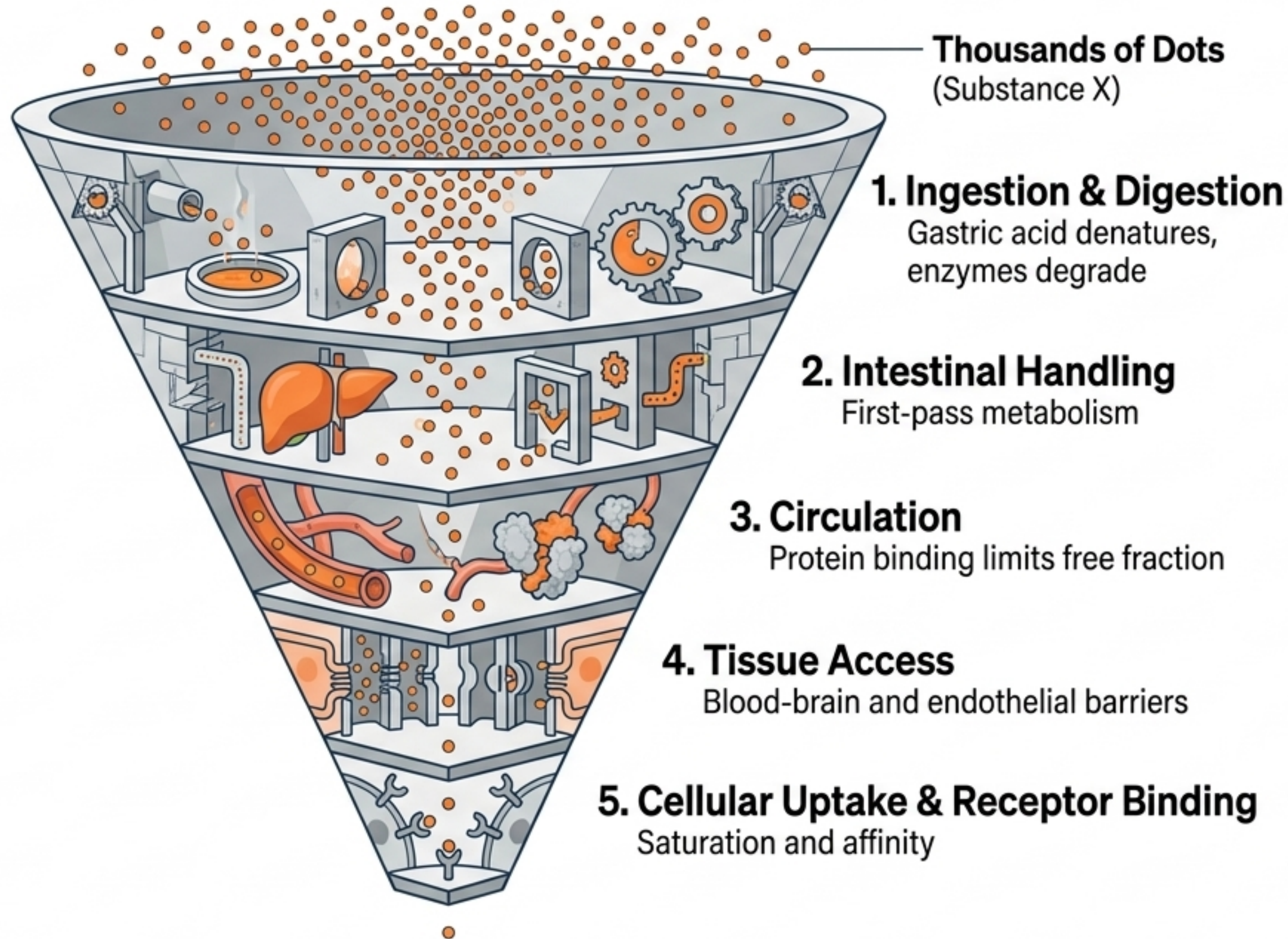


**You misread a lottery ticket as a winner
Your heart races.**

(A false interpretation of reality, but a 100% real physiological response).

**A real physical response does not prove the attributed external cause.
Expectation effects are biological events with informational causes.**

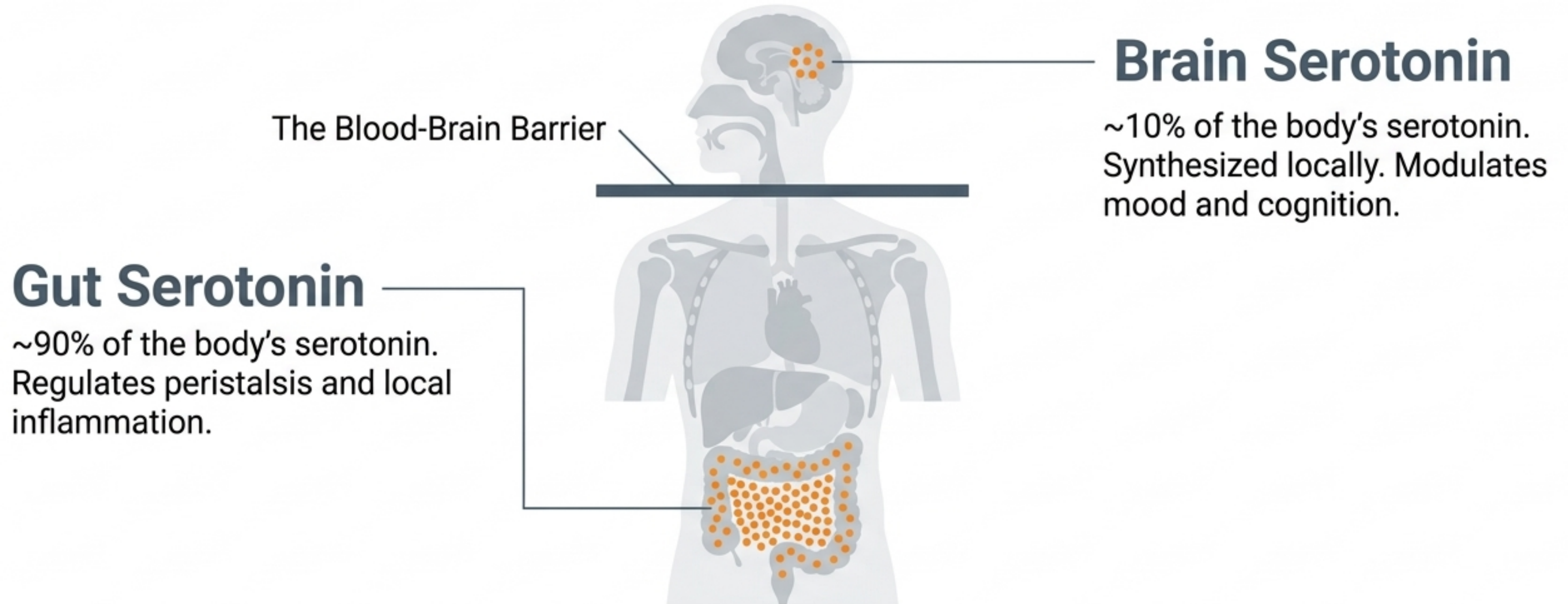
The Missing Middle of Biological Admissibility



'Food contains X, therefore eating it changes X in the body' deletes nearly all of physiology.

Between ingestion and behavior lies an obstacle course of clearance and regulation.

Compartmentalization: Biological Location Dictates Function

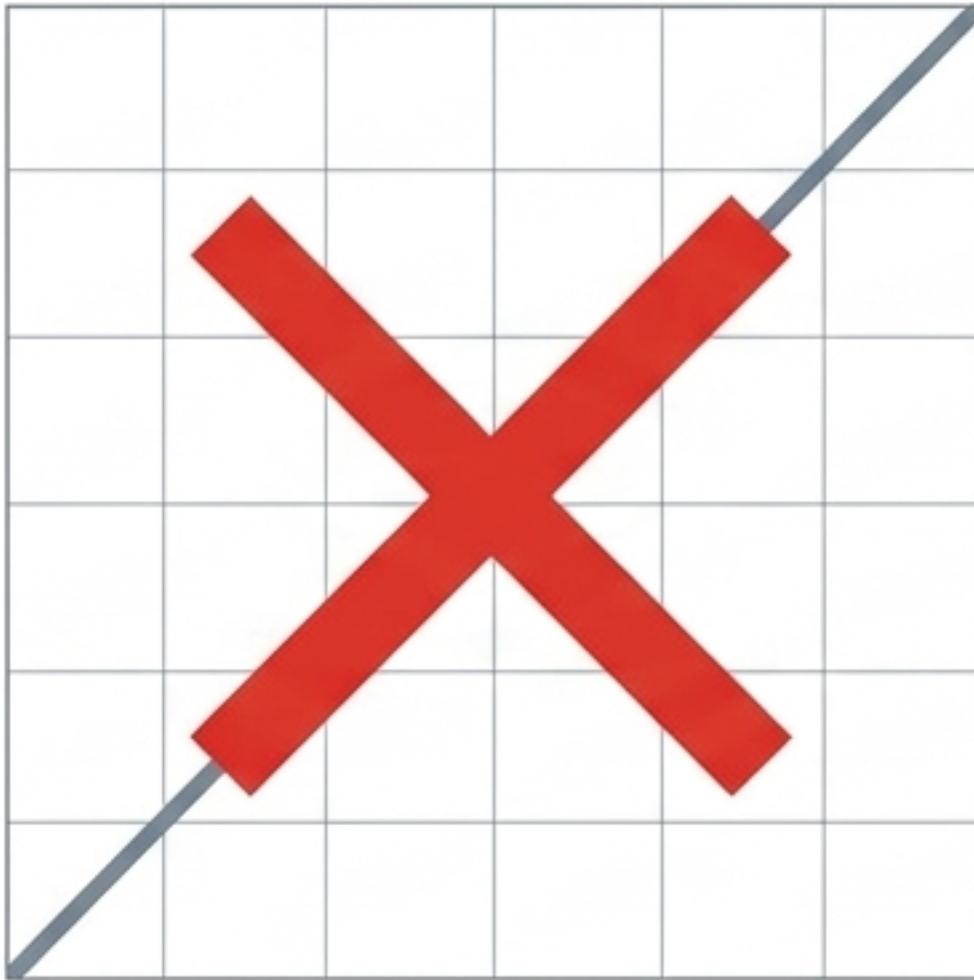


Most serotonin is not brain serotonin. The body does not respect the molecular name; it respects the physical compartment. Peripheral production does not mean systemic availability.

The Myth of the Monotonic Dose

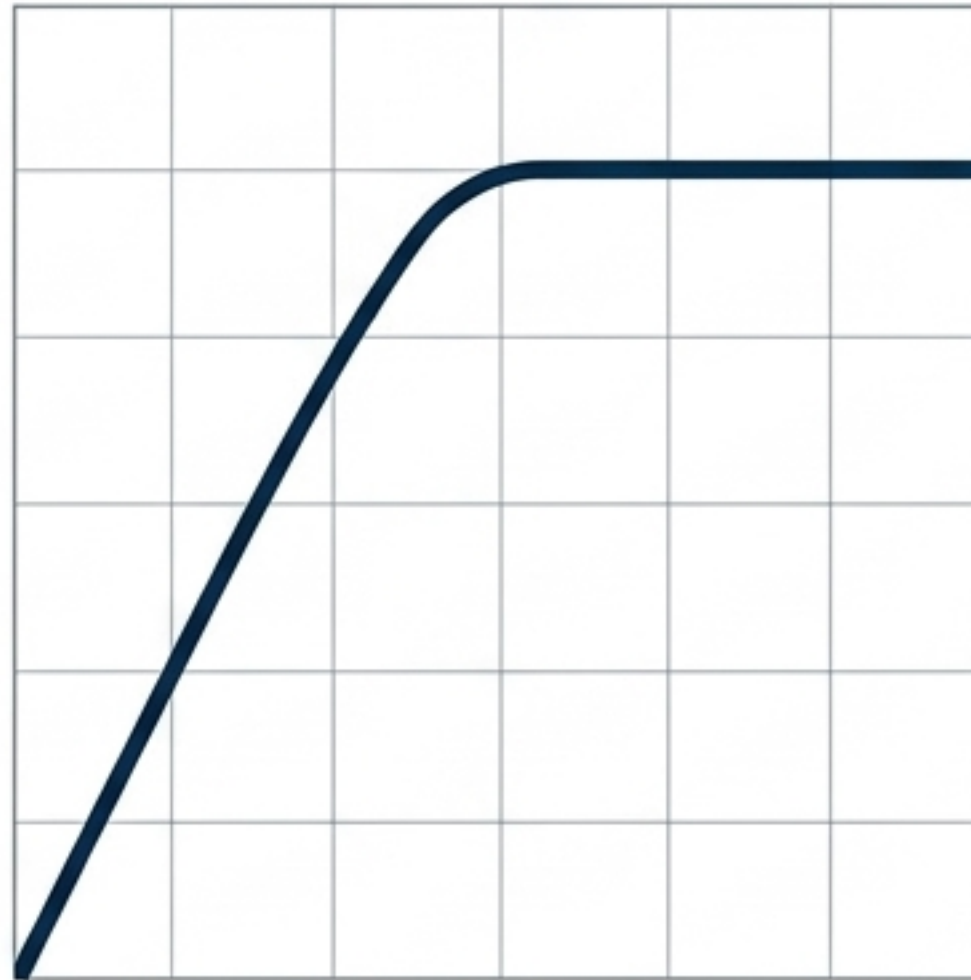
Folk Assumption

More is always more.



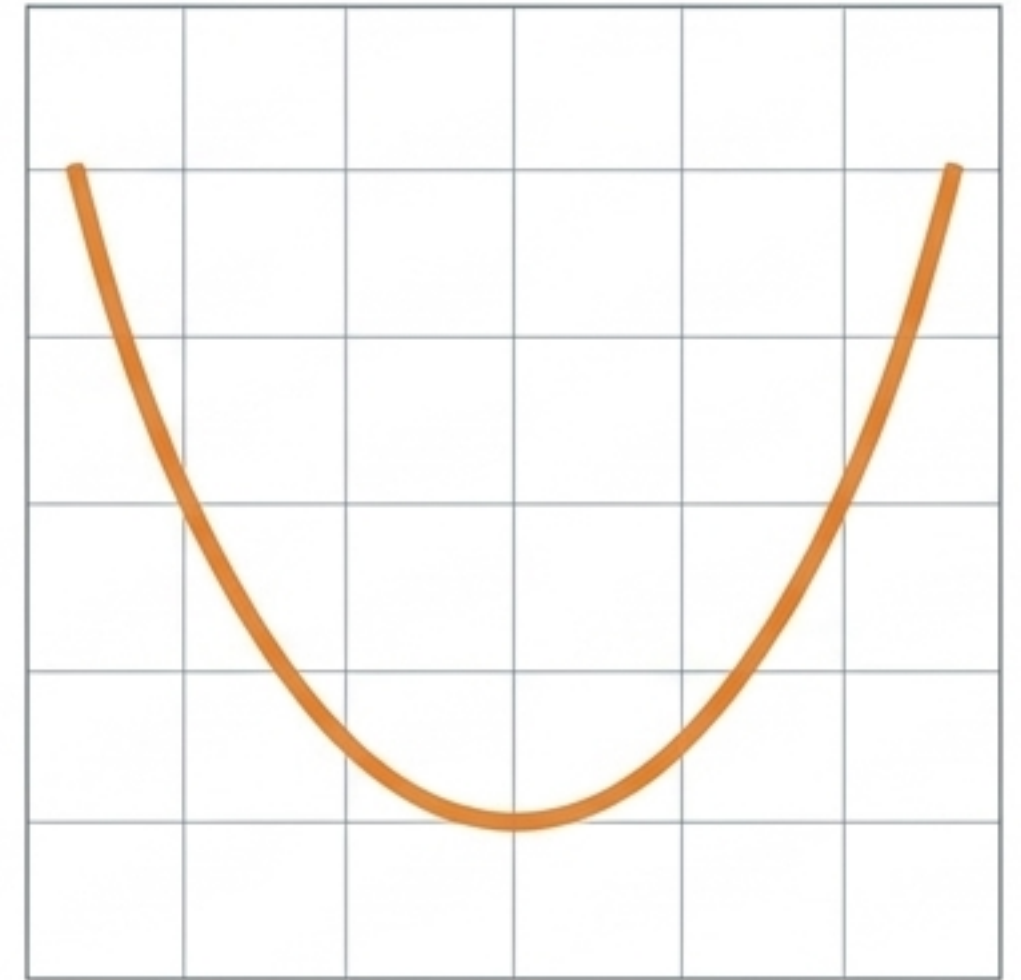
Receptor Saturation

Effects max out; adding more does nothing.



Endocrine / Non-Monotonic

Low doses stimulate; high doses trigger opposing loops.



The absence of an effect at a high dose does not guarantee safety at a low dose, and vice versa.

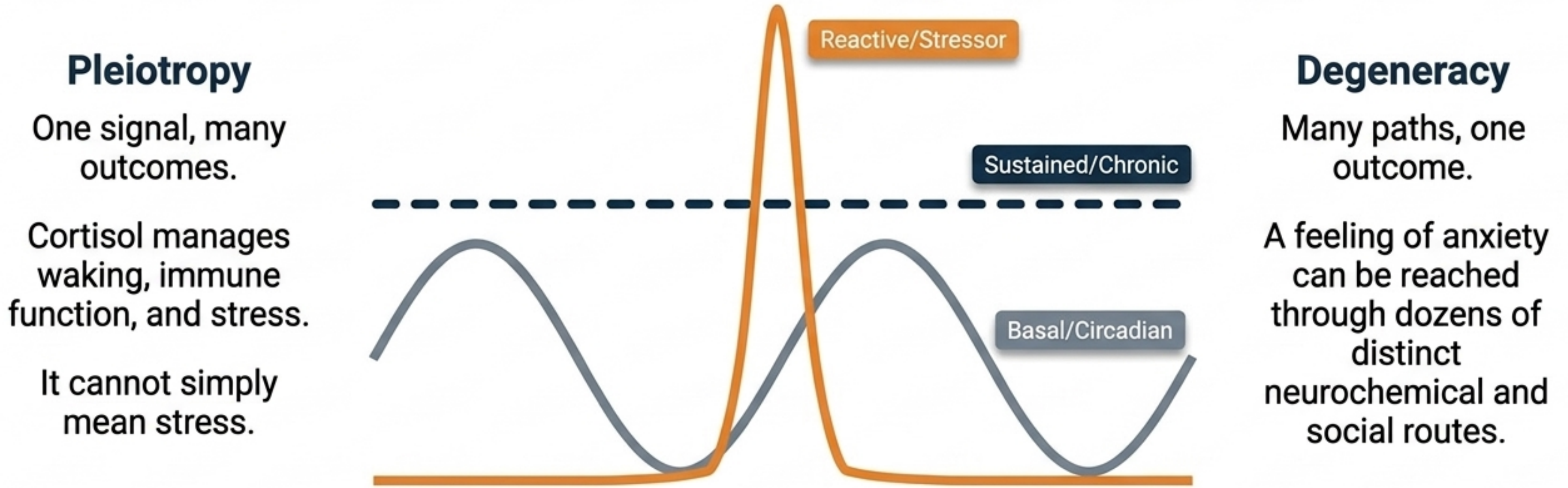
A Taxonomy of Causes

Role	Definition	Example
Vulnerability Factor	Raises probability/severity, but cannot trigger the outcome alone.	(e.g., Genetic predisposition)
Permissive Condition	Must be satisfied for other causes to operate.	(e.g., Adequate lighting to read)
Mediator	Transmits an effect from an upstream cause to a downstream outcome.	(e.g., Poor sleep transmitting the effect of stress)
Trigger	The proximate occasion that crosses a prepared threshold.	(e.g., A harsh word)
Correlated Marker	Tracks an outcome without being causally connected to it.	(e.g., A biomarker sharing an upstream cause)

Hormones act as Vulnerability Factors or Mediators. They are rarely the proximate Trigger.

The Cortisol Trap: Measurement ≠ Governance

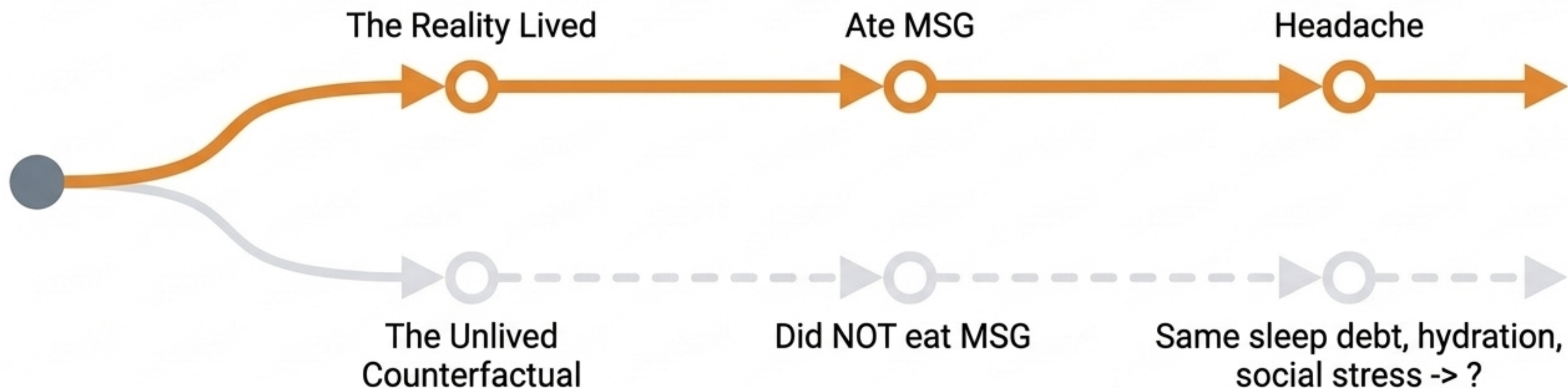
When someone says “my cortisol is high,” which line are they referring to?



A biomarker can be a downstream consequence or a feedback loop trying to regulate stress—not just the hidden author of it.

The Missing Counterfactual

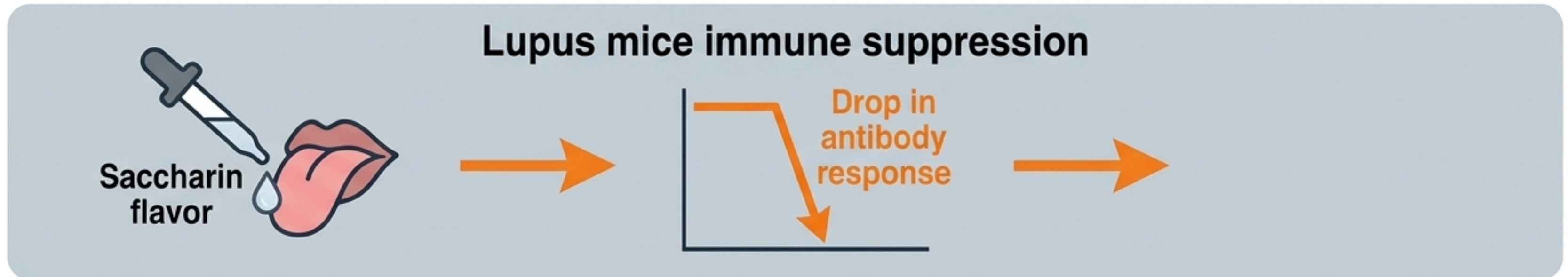
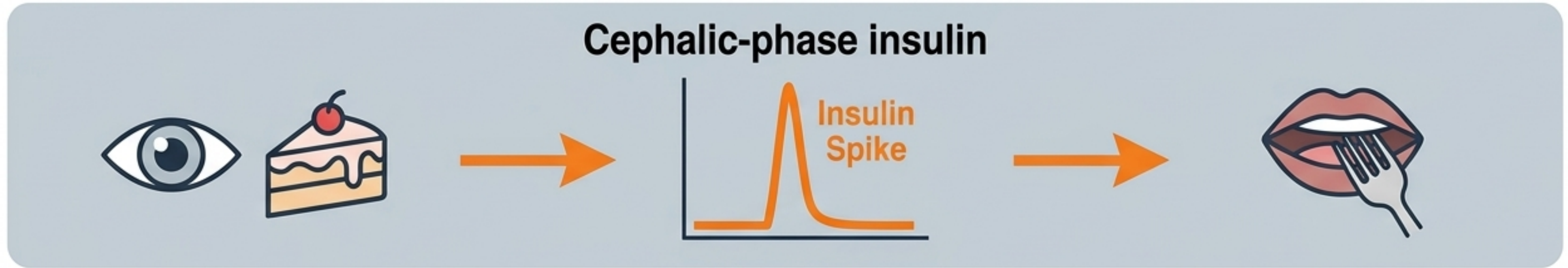
“I ate this and then developed a headache” accurately reports a sequence, but lacks a comparison. A single life follows one realized branch.



A lived sequence supplies temporal order, not its own comparison condition.

Causal methods (blinding, placebos, crossovers) are technologies designed to approximate this unavailable parallel universe.

Anticipatory Physiology: Information Before Exposure

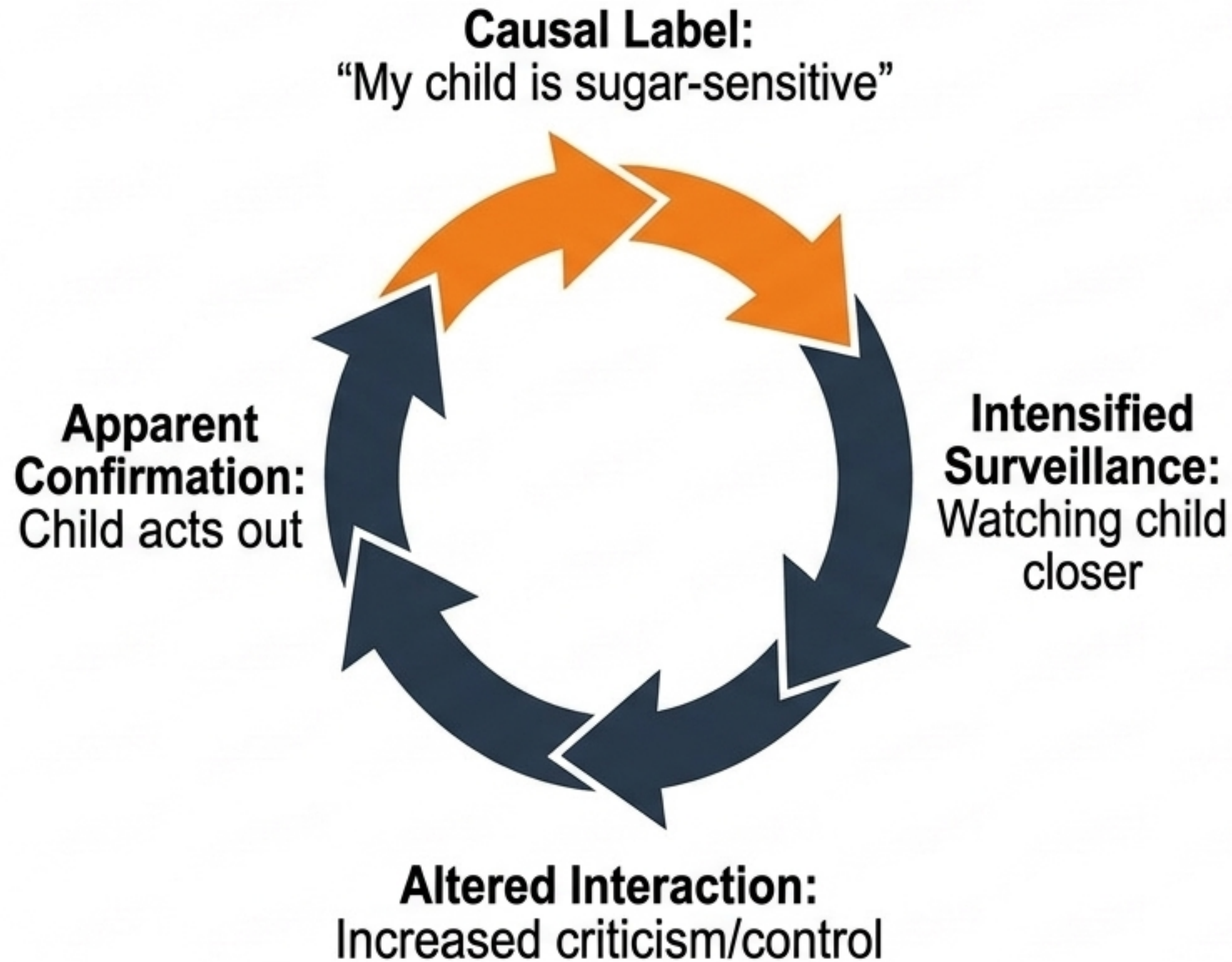


The body acts on evidence that an exposure is coming before the chemical reaches circulation.

In **classic** experiments, mice associated saccharin flavor with an immunosuppressive drug. Later, the flavor alone suppressed their immune system, extending survival in lupus-prone mice.

The biological response is frequently to the appraisal or conditioned cue, not the chemical substance.

Expectation as Social Biology



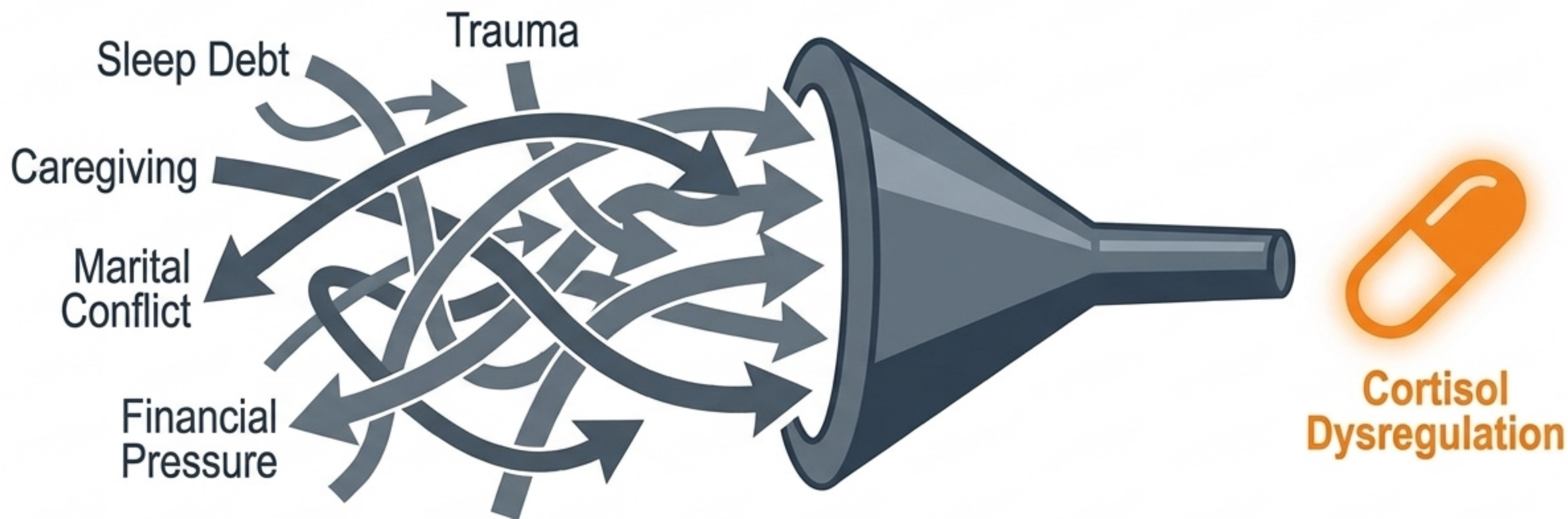
The Hoover/Milich Study (1994):

Mothers who were told their sons received a massive dose of sugar rated them as hyperactive and criticized them more often.

The Catch: All children received a sugar-free placebo.

Takeaway: A causal story changes the surveillance applied and the conduct directed toward a person. The chemical category becomes an **active** component of the present social cause.

Chemical Stories as Cognitive Strategies

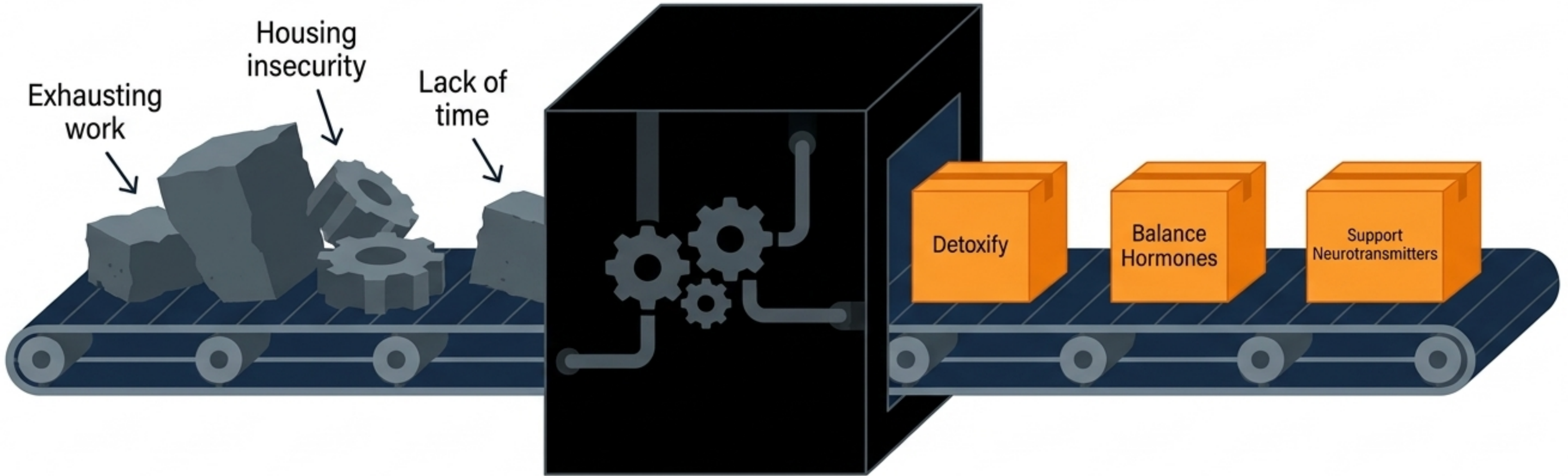


Why do false biological explanations persist? Because they successfully reduce causal entropy.

- **Identity:** Gives diffuse suffering a concrete boundary.
- **Externalization:** Protects the self by shifting agency ('My hormones did it').
- **Tractability:** Offers a defined, purchasable intervention.

Folk chemistry may misrepresent causation, but it is highly effective at organizing emotion

The Commercial Feedback Loop



Markets naturally select explanations that imply transactions.

It is incredibly difficult to package and sell solutions for caregiving burdens or abusive work environments. It is very easy to sell a supplement for "hormone imbalance."

Molecular explanations serve to individualize and privatize conditions whose largest causes are actually infrastructural.

They offer a defined, purchasable intervention for a complex, systemic problem, effectively shifting the agency and cost onto the individual.

The Blueprint: A Practical Standard for Chemical Claims

1

Specify Outcome

Define exact biological/behavioral change.

2

Establish Exposure

Ensure dose/route matches real-world use.

3

Trace Route

Map survival through barriers and digestion.

4

Measure Perturbation

Validate target receptor interaction.

5

Connect to Experience

Link downstream effect to behavior.

6

Construct Comparison

Demand counterfactuals (blinding).

7

Control Observer

Account for expectation loops.

8

Demand Stability

Require convergence across methods.

9

Ordinate Risk

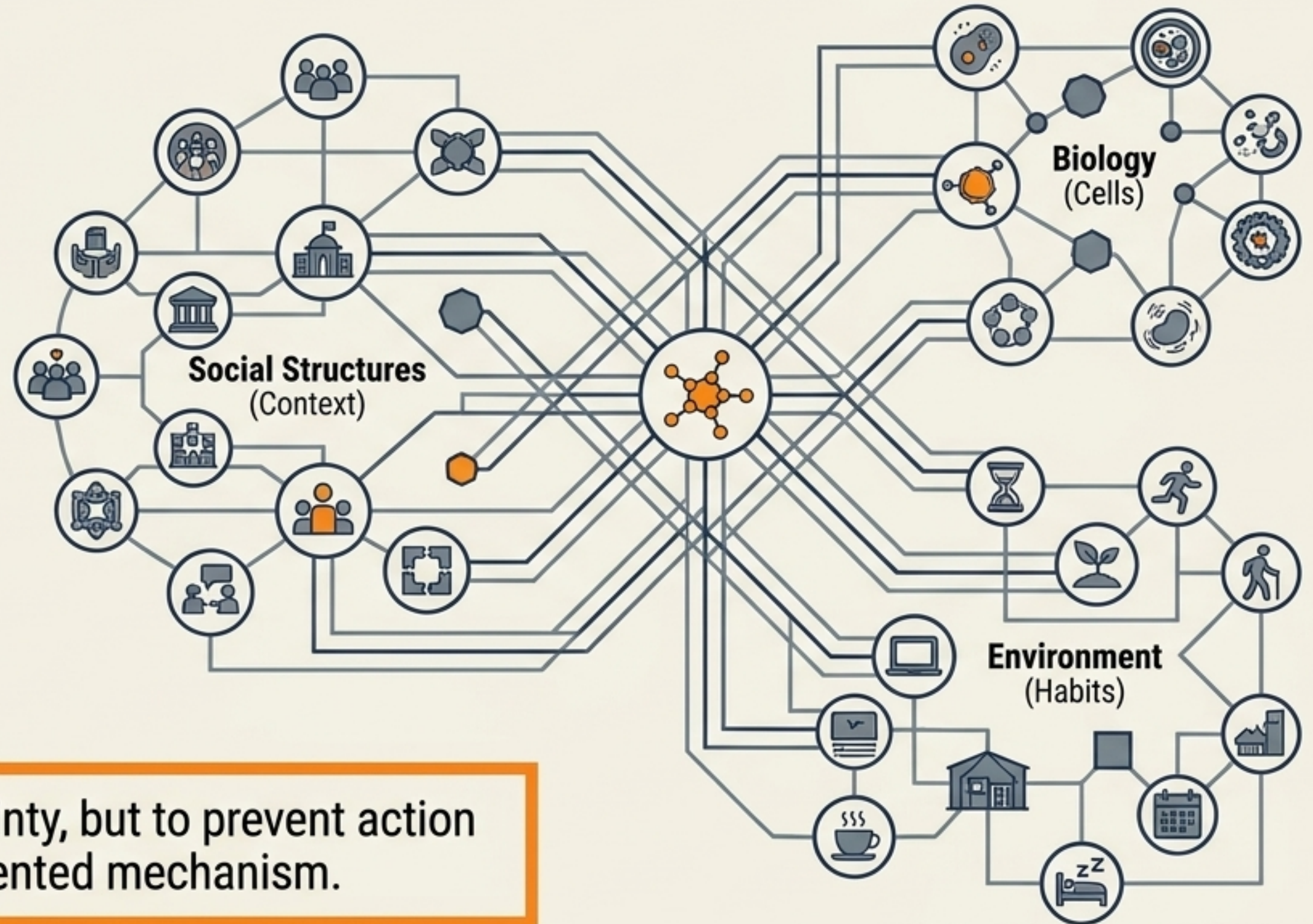
Weigh probability against competing life risks.

An Orientation, Not Just a Verdict

We must replace a false causal proposition with a new orientation.

A biological transition matters without dictating conduct.

Biology and social organization are not rivals; they are interacting parts of one causal system.



The goal is not to eliminate uncertainty, but to prevent action from hardening uncertainty into invented mechanism.